Photo to Printed Insert

Turning a $7.99 pistol case into a custom equipment case





The Mills Fleet Farm single pistol case ($7.99) ships full of foam. The goal: replace the foam with a 3D-printed insert shaped exactly to the gear.

OVERVIEW

The process at a glance

1
Photo
Object next to a printed ArUco marker

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2
Python script
Scale, perspective fix, trace

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3
DXF
Exact and 0.4 mm clearance outlines

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4
FreeCAD
Import, pad, pocket

>

5
Bambu Studio
Split to fit the plate

>

6
Print
Insert for the case

- A 3D-printed ArUco marker in the photo provides true scale.
- The script corrects for photos that are not perfectly in plane.
- It traces each outline and simplifies it into a clean polygon.
- DXF files import directly into any CAD program. In this case, FreeCAD.

STEP 1-3

Photograph and trace



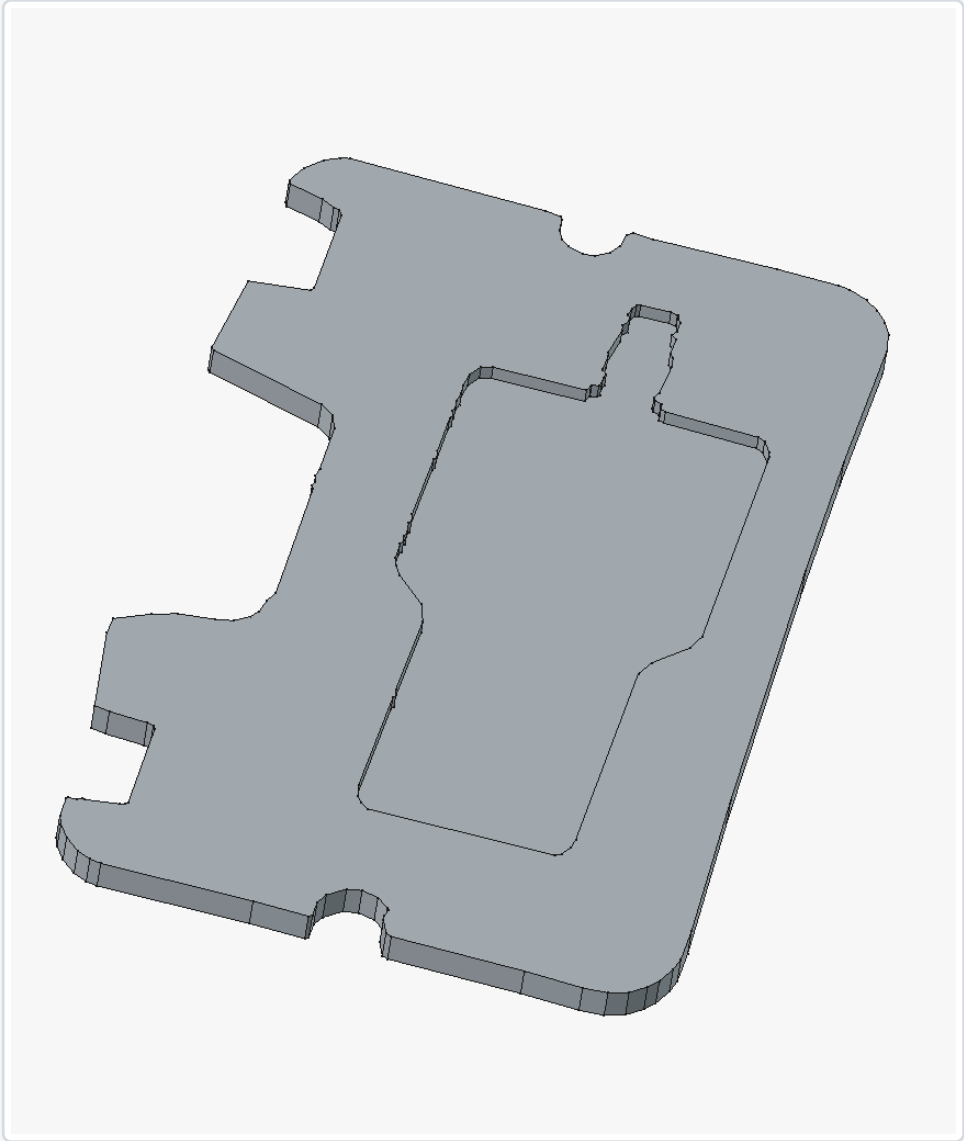


Foam piece (left) and RigExpert analyzer (right). Green is the traced object, blue is the detected marker. The marker sets the scale and its corners drive the perspective correction. The script writes an exact outline and a version with 0.4 mm printing clearance.

Analyzer outline: 96.9 x 241.3 mm exact | 97.7 x 242.1 mm with clearance

STEP 4

Cut the pocket in FreeCAD

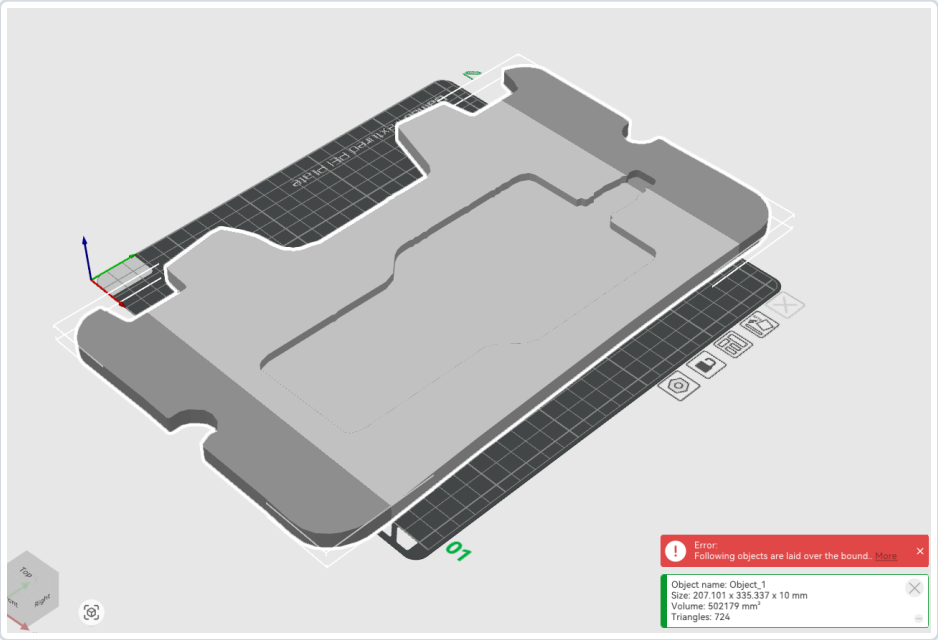


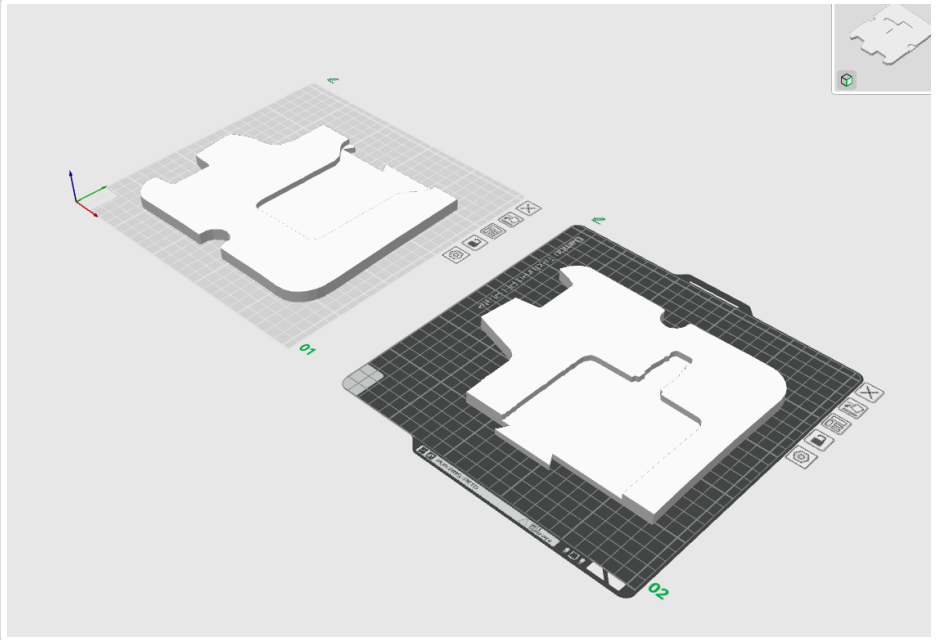
The foam piece outline becomes the plate. The analyzer outline is cut out of it as a pocket.

Both shapes come straight from the DXF files, already at true size, with no manual dimensioning.

STEP 5

Too big for one plate, so split it





Left: imported into Bambu Studio, the 207 x 335 x 10 mm insert exceeds the build plate. **Right:** the slicer cut tool splits it into two plates that fit.

SUMMARY

What comes out of it

One marker, one photo, one command per object

Scale and perspective come from the marker, so no ruler and no manual calibration.

True-scale DXF output

Exact and clearance versions, ready to import into any CAD program.

CAD to slicer without redrawing

FreeCAD for the pocket cut, Bambu Studio for splitting and printing.

Next: refine before printing

Smoother edges with a larger simplify value, then print the insert.